

Quick start to device development

V1.4

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1.Windows development

1.1. DLL development

For the windows development of C#,VC++,VB,VB.net , Delphi ,java , please references "[doc\English\windows\DLL Development Guide.doc](#)", before development .

1.1.1 Reader

1.1.1.1 HF Reader

1.1.1.1.1. Driver dependence library

Devices	RDType	Dll driver name	Remarks
RL15x/RL17x/RL86x/ TPAD10M	RL8000	rfidlib_drv_RL8000.dll	
RD201/M201/M202	M201	rfidlib_ANM201.dll	
RD51xx/RD242	RD5100	rfidlib_drv_RD5100.dll	
RD5200/RD3000/RD210 0/M22	RD5200	rfidlib_drv_RD5200.dll	1. The M22 module is compatible with the protocol of M201, if M201 has been developed before and the functions remains unchanged, it can continue to use M201 driver without modifying the software. 2. If you have developed the RD5100 before, and now use the RD2100/RD3000/RD5200, you can continue to use the RD5100 driver without modifying the software while maintaining the original functions

PBYTJ311/PBYTJ312/PB YTJ240/PBYTJ160/TPAD 20M/TPAD20L/RD212BL	These models belong to the all-in-one machine class, which is embedded with M201/M22 module. You can check whether the embedded module is M201 or M22 module through the R-Tool search interface, and develop the corresponding module according to the above module.
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1.1.1.1.2. API interface library

Driver	Description	doc
rfidlib_reader.dll	Required, Fundamental API	doc\English\windows\DLL API Description \ rfidlib_reader
rfidlib_aip_iso15693.dll	Option, API for ISO15693 protocol	doc\English\windows\DLL API Description \ rfidlib_aip_iso15693
rfidlib_aip_iso14443A.dll	Option dll , API for ISO14443A protocol	doc\English\windows\DLL API Description \ rfidlib_aip_iso14443A
rfidlib_aip_iso18000p3m3.dll	Option dll , API for ISO18000-3 mode3 protocol	doc\English\windows\d DLL API Description \ rfidlib_aip_iso18000p3m3

1.1.1.1.3. Default connection parameter of devices

Devices	Serialport	TCP	WIFI	Bluetooth
RL15x/RL17x/RL86x/TPAD10 M	Baudrate : 38400 Frame: 8E1	IP: 192.168.1.222 Port: 9900	/	/
RD201/M201/M202	Baudrate : 38400 Frame: 8E1	/	/	/
RD51xx/RD242	Baudrate : 38400 Frame: 8E1	IP: 192.168.1.222 Port: 9909	/	/
RD5200/RD3000/RD2100/M 22	Baudrate : 38400 Frame: 8E1	IP: 10.168.1.222 Port: 9909	/	/

1.1.1.1.4. Sample description

1.1.1.1.4.1. Inventory and operation tag memory

Program language	Demo path
C++	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\c++\TagAccess
C#	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\C#\TagAccess
Java	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\java\TagAccess
Delphi	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\delphi\TagAccess
VB6	1.samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\VB6\tagAccess 2.samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\VB6\MultipleProtocol
VB.net	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\VB.net\TagAccess

1.1.1.1.4.2. Set IO port output

Program language	Demo path
C++	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\c++\Setoutput
C#	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\C#\SetOutput_cs

1.1.1.1.4.3. RS485 sample

Program language	Demo path
C++	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\c++\RS485-Sample
C#	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\C#\RS485Sample_cs

1.1.1.1.4.4. Tag command transparent transceive operation

Program language	Demo path	Description
C#	(1)samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\C#\ISO15693TransparentTransceive	ISO15693 transparent transceive sample

	(2)samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\C#\ISO14443ATransparentTransceive	ISO14443A-3 transparent transceive sample, at present, only supported by RL86x reader
	(3)samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\C#\ISO14443Ap4_Transceive	ISO14443A-4 APDUtransparent transceive sample, at present, only supported by RL86x reader

1.1.1.1.4.5. Get input status

Program language	Demo path	Description
C#	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器 \C#\GetInputStatusSample	

1.1.1.2 UHF Reader

1.1.1.2.1. Driver dependence library

Deveices	RD Type	Driver name
ER60M2\UM201\URD5118\URD2004 \URD6116\URD8104\URD1101\UM1 101\UM1201\UM2101	UM200	rfidlib_drv_UM200.dll

1.1.1.2.2. API interface library

Driver	Description	doc
rfidlib_reader.dll	Required, foundation API	doc\English\windows\dll description\ rfidlib_reader_dll
rfidlib_aip_iso18000p6C.dll	Required , UHF EPC read/write API	doc\English\windows\dll description\ rfidlib_aip_iso18000p6C_ dll

1.1.1.2.3. Default connection parameter of devices

Devices	Serial port	TCP	WIFI	Bluetooth
UM201	Baudrate:115200 Frame:8N1			
URD5118	Baudrate:38400 Frame:8E1			

1.1.1.2.4. Samples description

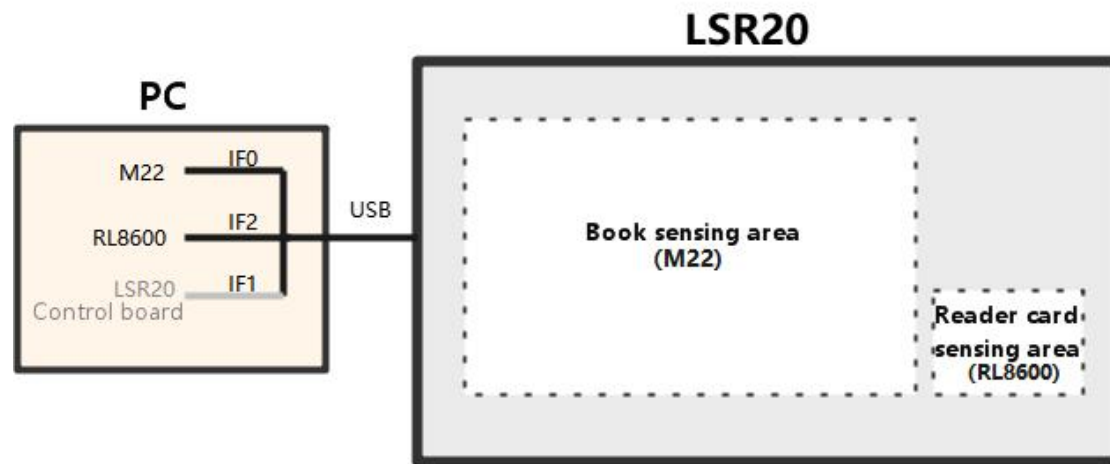
1.1.1.2.4.1. Inventory and tag memory operation

Program language	Demo path	description
C++	samples\rfid.sdk.windows\rfid_reader_UHF-超高频读卡器 \c++\WriteEPC	
C#	(1)samples\rfid.sdk.windows\rfid_reader_UHF-超高频读卡器 \C#\TagAccess (2)samples\rfid.sdk.windows\rfid_reader_UHF-超高频读卡器 \C#\writeEPC	
Java	samples\rfid.sdk.windows\rfid_reader_UHF-超高频读卡器 \java\WriteEPC	
VB6	samples\rfid.sdk.windows\rfid_reader_UHF-超高频读卡器 \VB6\tagAccess	

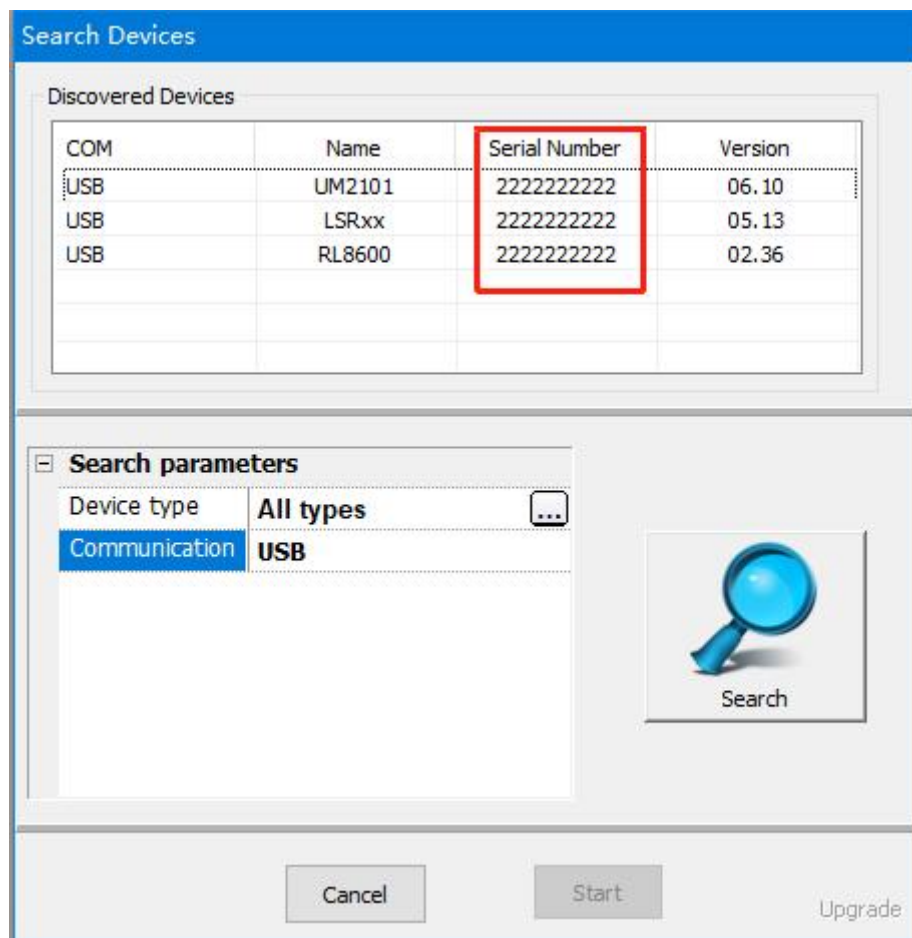
1.1.1.2.4.2. Set IO port output

Program language	Demo path
C++	samples\rfid.sdk.windows\rfid_reader_UHF-超高频读卡器\c++\Setoutput
C#	samples\rfid.sdk.windows\rfid_reader_UHF-超高频读卡器\C#\SetOutput_cs

1.1.1.3. LSR20



LSRXX use one USB port with three communication interface , one for M22/UM2101, one for LSRxx , and one for RL8600 . then you can search three device with same serial number , so you need to use addressed-mode to open the device .



The serial number address is defined as follows:

Serial number-0 is for M22/UM1201/UM2101 module

Serial number-1 is for Always LSR20 module (only for firmware upgrade)

Serial number-2 is for Always the RL8600 module

Serial number -2 is always RL8600 device

Standard	Tag type	Serial num	Read Cnt	Antenna No
ISO15693	Unknown	E00403500D116CBE	18	0
ISO15693	Unknown	E00403500D1173E9	18	0
ISO15693	Unknown	E00403500D11871D	18	0
ISO15693	Unknown	E00403500D1178A3	18	0
ISO15693	Unknown	E00403500D11644F	18	0
ISO15693	Unknown	E00403500D11894F	18	0

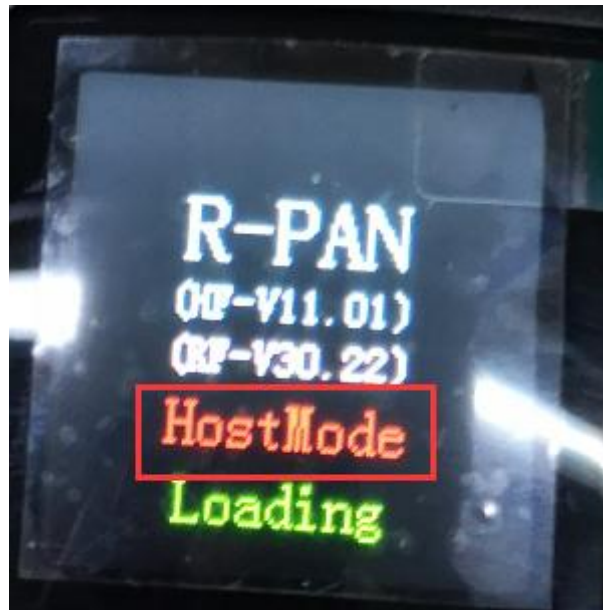
Total Count:
6

For the development of the M22, RL8600, UM201 reader modules, see "1.1.1.1 High-frequency Card Reader "and "1.1.1.2 UHF Card Reader" sections.

1.1.2 R-PAN/UR-PAN

1.1.2.1. Summary

R-PAN/UR-PAN support multiple work mode, include host control mode, scan mode and so on, work mode show in startup screen, as shown below:



Host control mode: In this mode, R-PAN/UR-PAN is in a passive mode, receiving commands from the host, and is developed in the same way as a universal card reader. For the development process, see "doc\English\windows\DLL Development Guide.doc". If there is a software that has developed M201 before, you can switch to R-PAN if you modify a small amount of code, because R-PAN is compatible with the protocol of M201.

Scanning mode: In this mode, R-PAN/UR-PAN automatically scans the book tag after start inventory. If configure read barcode, all the barcodes will save to ROM or RAM. The host obtains the records saved on R-PAN through commands. In this mode, the card reader inventory and read/write commands can't be delivered.

1.1.2.2. Driver dependence library

Device	RDType	dll driver name	Remarks
R-PAN	RPAN	rfidlib_drv_rpan.dll	
UR-PAN	UHF_RPAN	rfidlib_drv_rpan_uhf.dll	

1.1.2.3. API interface library

Drivers	Description
rfidlib_reader_dll	Required, device foundation API
rfidlib_aip_iso15693_dll	Option, API for ISO15693 protocol , use in host control mode of R-PAN
rfidlib_aip_iso18000p6C_dll	option , UHF EPC read/write API, use in host control mode of UR-PAN

1.1.2.4. Default connection parameter of devices

Devices	USB HID	WIFI	Bluetooth
R-PAN	/	AP:12345678	/
UR-PAN	/	AP:12345678	/

1.1.2.5. Sample description

1.1.2.5.1. Host mode

R-PAN: reference the sample of TagAccess on "HF Reader"

UR-PAN: reference the sample of TagAccess on "UHF Reader"

1.1.2.5.2. Scan mode

R-PAN:

Program language	Active scan mode/Library mode
C++	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器 \c++\BufferModeSample
C#	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器 \C#\BufferModeSample
Java	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器\java\BufferMode
Delphi	samples\rfid.sdk.windows\rfid_reader_HF-高频读卡器 \delphi\BufferModeSample

UR-PAN:

Program language	Active scan mode/Library mode
C#	samples\rfid.sdk.windows\rfid_reader_UHF-超高频读卡器 \C#\BufferModeSample

1.1.3 Library safety gate

1.1.3.1. Driver dependence library

Devices	RDType	Driver name
LSG405/LSG606/LSG406/LSG410	LSG405	rfidlib_LSG405.dll
The following library safety gate control center: LSG428/LSG422/LSG618/LSG620/LSG619/LSG701/LSG429	LSGControlCenter	rfidlib_drv_LSGControlCenter.dll

1.1.3.2. API interface library

Library name	Description
rfidlib_reader_dll	Required, operation API for safety gate. For more information about APIs, see the "rfidlib_reader.doc".

1.1.3.3. Default connection parameter of devices

Devices	Serial port	TCP	WIFI	Bluetooth
LSG405/LSG606/LSG406/LSG410		IP:192.168.1.222 Port:6012		
LSG428/LSG618/LSG620/LSG619/LSG701	Baudrate:38400 Frame:8E1	IP:10.168.1.222 Port:6012	AP password: 12345678	

1.1.3.4. Sample description

1.1.3.4.1. LSG405/LSG606/LSG406/LSG410 Development

1.1.3.4.1.1. Obtain alarm record、 people flow info

Program language	References sample
C#	samples\rfid.sdk.windows\rfid_library_security_gate-图书馆安全门 \C#\LSG_Csharp_demo
Java	samples\rfid.sdk.windows\rfid_library_security_gate-图书馆安全门 \java\LSG406
Delphi	samples\rfid.sdk.windows\rfid_library_security_gate-图书馆安全门 \delphi

1.1.3.4.1.2. Set alarm output

Program language	References sample
C#	samples\rfid.sdk.windows\rfid_library_security_gate-图书馆安全门 \C#\SetOutput_cs
Java	samples\rfid.sdk.windows\rfid_library_security_gate-图书馆安全门 \java\LSG406

1.1.3.4.2. LSG42x/LSG61x application development

Program language	References sample
C#	samples\rfid.sdk.windows\rfid_library_security_gate-图书馆安全门 \C#\LSG_Control_Center_Csharp_demo
Java	samples\rfid.sdk.windows\rfid_library_security_gate-图书馆安全门 \java\LSGControlCenter(Eclipse)/(LSGControlCetner_IDEA)
C++	samples\rfid.sdk.windows\rfid_library_security_gate-图书馆安全门 \mfc\LSGControlCenter

1.1.3.4.3. LSG42x/LSG61x notification mode development

Program language	References sample
C#	samples\rfid.sdk.windows\rfid_library_security_gate-图书馆安全门 \C#\TestServer

1.1.4 Meeting gate

1.1.4.1. Driver dependence library

Deveice	RDType	Driver
G302/G303	G302	rfidlib_G302.dll

1.1.4.2. API interface library

Library name	Description
rfidlib_reader_dll	Required,operation API for meeting gate. For more information about APIs, see the “rfidlib_reader.doc”.

1.1.4.3. Default connection parameter of devices

Devices	Serial port	TCP	WIFI	Bluetooth
LSG301/LSG302	Baudrate:38400 Frame:8E1	IP:192.168.1.222 Port:6012		

1.1.4.4. Sample description

Program language	References sample
C#	samples\rfid.sdk.windows\rfid_meeting_gate-会议签到门 \c#\MeetingGateAsst
Java	samples\rfid.sdk.windows\rfid_meeting_gate-会议签到门\java
C++	samples\rfid.sdk.windows\rfid_meeting_gate-会议签到门 \mfc\MeetingGateAsst
Delphi	samples\rfid.sdk.windows\rfid_meeting_gate-会议签到门 \delphi\meeting_gate_delphi_demo1

1.1.5 LCC-Smart bookshelf middle-ware

1.1.5.1. Relevant document

Doc	Description
doc\English\windows\ LCC-智能书架中间件开发指南\ LAS_Server_For_LCC 使用说明文档 -V1.0.1	1、dll and jar, description for Call process 2、Detailed interface development document

1.1.5.2. Dll and sample description

Program language		Description
C#	samples\rfid.sdk.windows\ LCC-智能书架中间件开发包 \C#\DLL	Directory for dll
	samples\rfid.sdk.windows\ LCC-智能书架中间件开发包 \C#\Test_LAS-Demo	Demo source code
	samples\rfid.sdk.windows\ LCC-智能书架中间件开发包 \C#\Test_LAS-Release	Directory for exe
Java	samples\rfid.sdk.windows\ LCC-智能书架中间件开发包 \java\ LAS_Server_For_LCC.jar	Jar package
	samples\rfid.sdk.windows\ LCC-智能书架中间件开发包 \java\ RunJar-Demo.bat	Run Jar package in demo mode
	samples\rfid.sdk.windows\ LCC-智能书架中间件开发包 \java\ RunJar-Test.bat	Run Jar package in test demonstration mode

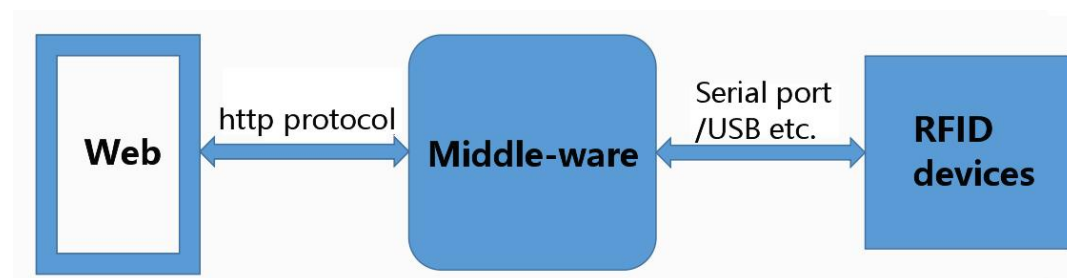
1.2. Web development

1.2.1. Use Active X plug-ins

The Active X only support windows IE. For details, see ["doc\English\windows\activeX web development guide.doc"](#).

1.2.2. Use middleware

To interact with the device in a non-IE browser environment, web middleware must be installed on the client first . The communication process is as follows:



At present, the function of the middleware is relatively simple, only support simple label reading and writing function, specific development instructions, please see ["doc\English\windows\activeX web development guide.doc"](#).

2. Android develop

2.1. Reader

2.1.1. HF Reader

2.1.1.1. Devices driver description

Device	RDType	Remarks
RL15x/RL17x/RL86x/TPAD10M	RL8000	
RD201/M201/M202	RD201	
RD51xx/RD242	RD5100	
RD5200/RD3000/RD2100/M22	RD5200	The M22 module is compatible with the protocol of M201, and if M201 has been developed before and the function remains unchanged, it can continue to use M201 without modifying the software.
R-PAN	RPAN	
PBYTJ311/PBYTJ312/PBYTJ240/PBYTJ160/TPAD20M/TPAD20L/RD212BL	These model numbers belong to the all-in-one machine class, and the M201/M22 module is embedded inside. You can check whether the M201 or M22 module is embedded through the R-Tool search interface. Then according to the above module corresponding development.	

2.1.1.2. Default connection parameter of devices

Devices	Serial port	TCP	WIFI	Bluetooth
RL15x/RL17x/RL86x/TPAD10M	Baudrate:38400 Frame:8E1	IP: 192.168.1.222 Port: 9900		
RD201/M201/M202	Baudrate:38400 Frame:8E1			
RD51xx/RD242	Baudrate:38400 Frame:8E1	IP: 192.168.1.222 port: 9909		
RD5200/RD3000/RD2100/M22	Baudrate:38400 Frame:8E1	IP: 10.168.1.222 port: 9909		
R-PAN				

2.1.1.3. Relevant API documentation

Doc	Description
doc\English\android\ DLL API Description\ADReaderInterface.doc	Devices foundation API
doc\English\android\DLL API Description\ISO15693Interface.doc	ISO15693 API
doc\English\android\DLL API Description\ISO14443AInterface.doc	ISO14443A API

2.1.1.4. Samples description

Develop tools	Reference sample
Android Studio	samples\rfid.sdk.android\rfid_reader_HF-高频读卡器\android studio \HF_TagAccess

2.1.2 UHF Reader

2.1.2.1. Devices driver

Devices	RDType
ER60M2\UM201\URD5118\URD2004\URD6116\URD8104 \URD1101\UM1101\UM1201\UM2101	UM200
UR-PAN	UR-PAN

2.1.2.2. Default connection parameter of devices

Devices	Serial port	TCP	WIFI	Bluetooth
ER60M2\URD5118\URD2004	Baudrate:38400 Frame:8E1			
UM201	Baudrate:115200 Frame:8N1			
URD6116\URD8104	Baudrate:38400 Frame:8E1	IP:192.168.1.222 Port:9909		
UM1101\UM1201\UM2101	Baudrate:115200 Frame:8N1			
UR-PAN				

2.1.2.3. Relevant API documentation

Doc	Description
doc\English\android\DLL API Description\ADReaderInterface.doc	Device foundation API
doc\English\android\DLL API Description\ISO18000p6CInterface.doc	ISO18000-6C API

2.1.2.4. Common Sample description

Development tools	References sample
Android Studio	samples\rfid.sdk.android\rfid_reader_UHF-超高频读卡器 \UHF_TagAccess

2.2. Library safety gate

2.2.1. Devices driver type

Devices	RDType
LSG405/LSG606	LSG405
Control center for Safety gate as below: LSG428/LSG422/LSG618/LSG620/LSG619/LSG701/LSG429	LSGControlCenter

2.2.2. Default connection parameter of devices

Devices	Serial port	TCP	WIFI	Bluetooth
LSG405/LSG606		IP:192.168.1.222 Port:6012		
LSG428/LSG618/LSG620/LSG619/ LSG701	Baudrate:38400 Frame:8E1	IP:10.168.1.222 Port:6012	AP password: 12345678	

2.2.3. Relevant API documentation

Document	Description
doc\English\android\DLL API Description \ADReaderInterface.doc	API for safety gate, references the part of ADReaderInterface.doc

2.2.4. Sample description

2.2.4.1. LSG405/LSG606

Development tools	References sample
Android Studio	samples\rfid.sdk.android\rfid_library_security_gate-图书馆安全门 \LSGDemoAS

2.2.4.2. LSG428/LSG619

Development tools	References sample
Android Studio	samples\rfid.sdk.android\rfid_library_security_gate-图书馆安全门 \LSGControlCenterAS

2.2.4.3. LSG422/LSG619 development of message notice mode

Development tools	References sample
Android Studio	samples\rfid.sdk.android\rfid_library_security_gate-图书馆安全门 \TestServerAS

2.3. Android handheld phone

The development process of the handheld device is consistent with the development process of the ordinary card reader Android, the only difference is that the handheld device needs to call the power control interface to provide power on and off service (handheld device development = card reader development + power on control);

The card reader is developed according to the RDType of the handheld computer, refer to the section "2.1 Reader " above.

The power-on control calls the power-on control function according to the handheldType

2.3.1. Power control function

Definition	public int HandheldPowerController(int handheldType, boolean swithType)		
Description	Handheld device power controller.		
Parameter	Name	Type	Remark
	handsetType	int	There are different types of handhelds 【0 AH2210, 1 AH1113】
	swithType	boolean	【true powerOn, false powerOff】
Return	0: Success, Not 0: Failure		
Notes			

2.3.2. Equipment parameter description

By default, handheld devices are connected by serial port. The default parameters of serial port connection and handheld type parameters are defined as follows:

device	RDType	handheldType	Default connection parameter
AH2210/AH2212/	RD5200	AH2210	Baudrate: 38400 Frame: 8E1
AH1112/AH2210L/AH1212	RL8000	AH2210	Baudrate: 38400 Frame: 8E1
UAH2210/UAH2212	UM200	AH2210	Baudrate: 115200 Frame: 8N1
AH1113	RL8000	AH1113	Baudrate: 38400 Frame: 8E1

3. Guide of Assembly product development

3.1. samples

3.1.1. Bookcase sample

Development tools	References sample
Android Studio	samples\rfid.sdk.android\rfid_bookcase-书柜\BookCaseAS
Visual Studio(C#)	samples\rfid.sdk.windows\rfid_bookcase-书柜

3.2. Bookcase

3.2.1. Book reservation bookcase (SRC101B)

Module	Model number	Guide of development		Communication type
ID card		references the part of "3.1.1"		USB
Fingerprint		references the part of "3.1.1"		USB
Lock		references the part of "3.1.1"		COM1(ttys0)
Light		references the part of "3.1.1"		COM2(ttys1)
HF reader	RD5212-F11	windows	references the part of "1.1.1.1"	TCP
		android	references the part of "2.1.1"	
Reader card	RL86S	windows	references the part of "1.1.1.1"	COM3(ttys2)
		android	references the part of "2.1.1"	
LED		references the part of "3.1.1"		COM4(ttys3)

3.2.2. MINI bookcase(SSM201B)

Module	Model number	Guide of development		Communication type
ID card		references the part of "3.1.1"		USB
Fingerprint		references the part of "3.1.1"		USB
Lock		references the part of "3.1.1"		COM1(ttys0)
Light		references the part of "3.1.1"		COM2(ttys1)
HF reader	RD5212-F11	windows	references the part of "1.1.1.1"	COM4(ttys3)
		android	references the part of "2.1.1"	
UHF reader	URD5118	windows	references the part of "1.1.1.2"	COM4(ttys3)
		android	references the part of "2.1.2"	
Reader card	RL86S	windows	references the part of "1.1.1.1"	COM3(ttys2)
		android	references the part of "2.1.1"	

3.2.3. Micro library bookcase (SSP301B)

Module	Model number	Guide of development		Communication type
ID card		references the part of "3.1.1"		USB
Fingerprint		references the part of "3.1.1"		USB
Lock		references the part of "3.1.1"		COM1(ttys0)
Light		references the part of "3.1.1"		COM2(ttys1)
HF reader	RD5212-F11	windows	references the part of "1.1.1.1"	TCP
		android	references the part of "2.1.1"	
UHF reader	URD5124	windows	references the part of "1.1.1.2"	TCP
		android	references the part of "2.1.2"	
Reader card	RL86S	windows	references the part of "1.1.1.1"	COM3(ttys2)
		android	references the part of "2.1.1"	
LED		references the part of "3.1.1"		COM4(ttys3)

3.2.4. Book Corner(SSM301B)

Module	Model number	Guide of development		Communication type
ID card		references the part of "3.1.1"		USB
Fingerprint		references the part of "3.1.1"		USB
Lock		references the part of "3.1.1"		ttys2
Light		references the part of "3.1.1"		ttyusb0
HF reader	RD2104-F11	windows	*	*
		android	references the part of "2.1.1"	ttys3
UHF reader	URD5118	windows	*	*
		android	references the part of "2.1.2"	ttys3
Reader card	RL86S	windows	*	*
		android	references the part of "2.1.1"	ttys1

3.3. Self-service borrowing and returning machine

3.3.1. Vertical self-service machine(SSK100)

Module	Model number	Guide of development		Communication type
ID card		references the part of "3.1.1"		USB
Fingerprint		references the part of "3.1.1"		USB
Printer		references the part of "3.1.1"		COM4(ttys3)
HF reader	PBYTJ311	windows	references the part of "1.1.1.1"	COM2 (ttys1)
		android	references the part of "2.1.1"	
UHF reader	ER60M2	windows	references the part of "1.1.1.2"	COM2 (ttys1)
		android	references the part of "2.1.2"	
Reader card	RL86S	windows	references the part of "1.1.1.1"	COM3 (ttys2)
		android	references the part of "2.1.1"	

3.3.2. Desktop self-service machine(SSK201)

Module	Model number	Guide of development		Communication type
ID card		references the part of "3.1.1"		USB
Printer		references the part of "3.1.1"		COM2 (ttys1)
HF reader	PBYTJ311	windows	References the part of "1.1.1.1"	COM1(ttys0)
		android	references the part of "2.1.1"	
UHF reader	ER60M2	windows	references the part of "1.1.1.2"	COM1(ttys0)
		android	references the part of "2.1.2"	
Reader card	RL863	windows	references the part of "1.1.1.1"	USB
		android	references the part of "2.1.1"	

3.3.3. K12 vertical self-service machine(SSK301)

Module	Model number	Guide of development		Communication type
ID card		references the part of "3.1.1"		USB
Fingerprint		references the part of "3.1.1"		USB
Printer		references the part of "3.1.1"		COM2 (ttys1)
HF reader	PBYTJ311	windows	references the part of "1.1.1.1"	COM1 (ttys0)
		android	references the part of "2.1.1"	
UHF reader	ER60M2	windows	references the part of "1.1.1.2"	COM1 (ttys0)
		android	references the part of "2.1.2"	
Reader card	RL863	windows	references the part of "1.1.1.1"	USB
		android	references the part of "2.1.1"	

3.3.4. Children's self-service machine(SSK501)

Module	Model number	Guide of development		Communication type
ID card		references the part of "3.1.1"		COM3 (ttys2)
Fingerprint		references the part of "3.1.1"		USB
Printer		references the part of "3.1.1"		COM4 (ttys3)
HF reader	PBYTJ311	windows	references the part of "1.1.1.1"	COM2 (ttys1)
		android	references the part of "2.1.1"	
UHF reader	ER60M2	windows	references the part of "1.1.1.2"	COM2 (ttys1)
		android	references the part of "2.1.2"	
Reader card	RL863	windows	references the part of "1.1.1.1"	USB
		android	references the part of "2.1.1"	

3.4. Intelligent inventory vehicle(MIT101)

Module	Model number	Guide of development		Communication type
HF reader	RD2104-F11	windows	References part of "1.1.1.1"	COM1(ttys1)
		android	References part of "2.1.1"	